# DATA ANALYTICS AND VISUALIZATION USING APEX

HACKATHON – Level-2 Template (Data Implementation & Interactive Analytics)

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| STUDENT NAME | M.bala |
| YEAR: | 20226 |
| COLLEGE CODE: | 105 |
| COLLEGE NAME: | J.P.college of arts and science |
| DEPARTMENT: | B.sc Computer science |
| SEMESTER: | IV |
| ROLL NO: | 2408105180211107 |
| NMID: | 07e83b6727898299f7b4fb88377f4185@nm2026.in |
| DATE: | 07.04.26 |
| TITLE: | EduPulse Insights |

## 1. Objective

The primary goal of Level-2 is to transform the conceptual data model designed in Level-1 into a fully functional Data Analytics and Visualization System using Oracle APEX. This phase focuses on implementing database tables, populating them with meaningful educational data, and creating an interactive dashboard that provides real-time insights into student performance, attendance trends, and batch-wise comparisons.

## 2. Problem Focus

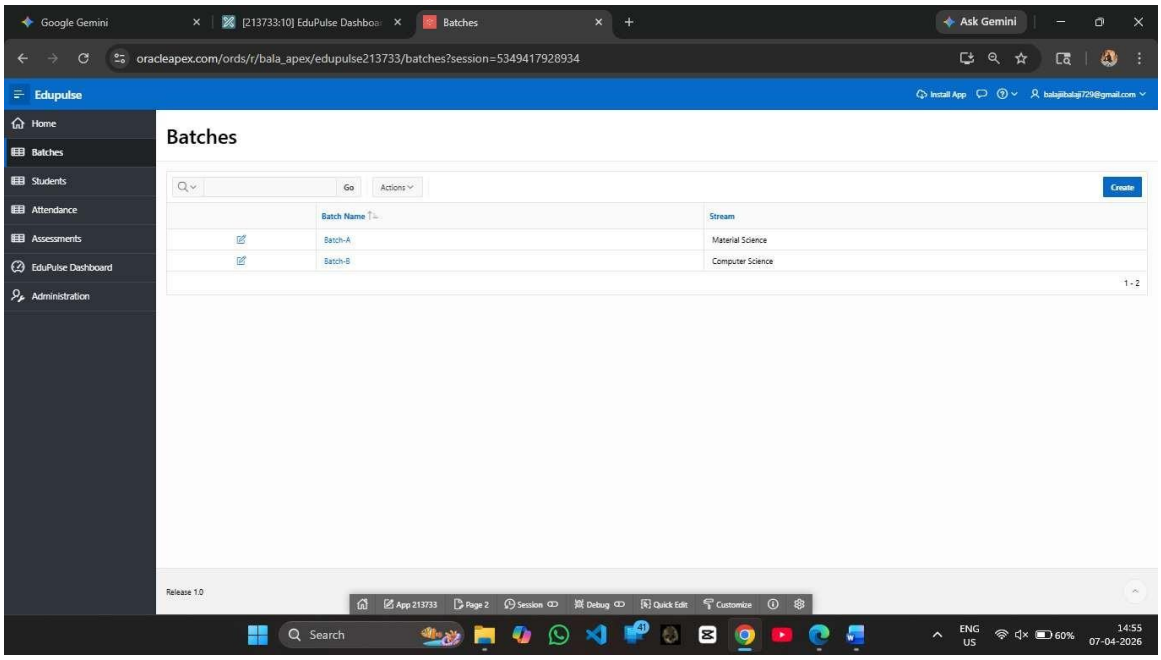
In Level-1, the focus was purely on architectural design. Level-2 addresses the challenge of converting raw, static data into actionable intelligence. The transition involves building a robust back-end to store academic records and

a dynamic front-end that allows educators to identify low-performing students and attendance gaps through interactive reports and high-fidelity dashboards.

3. Data Model Implementation

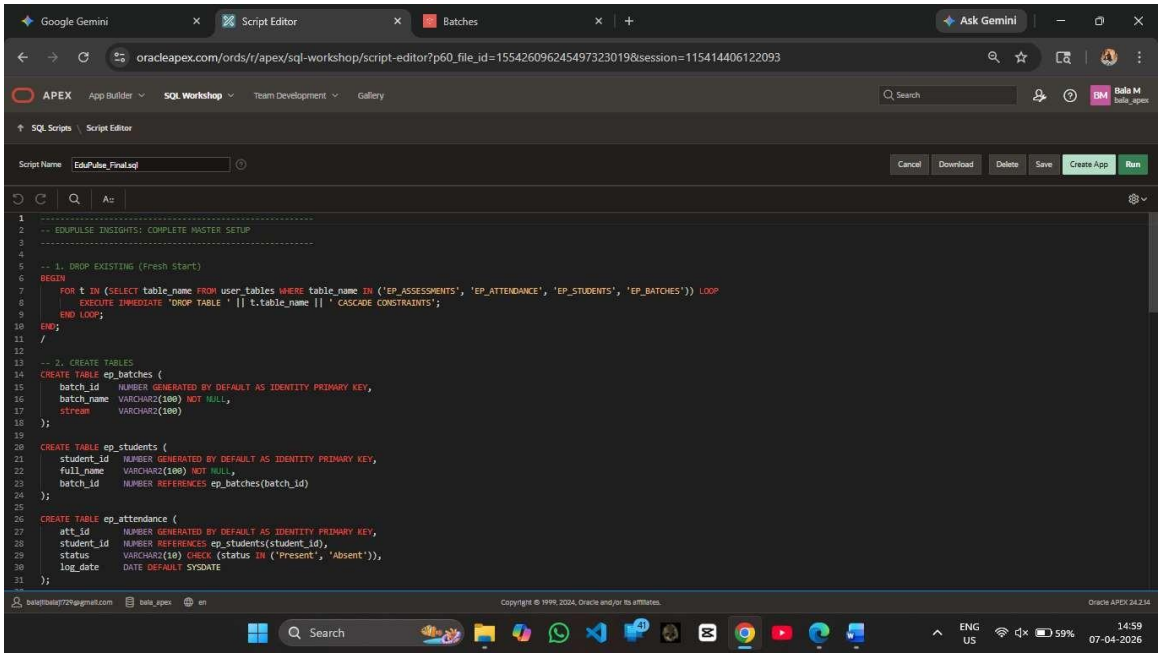
We implemented a relational database structure consisting of four core tables to ensure data integrity and logical relationships:

- EP_BATCHES: Stores batch names and academic streams (e.g., Material Science) using NUMBER for IDs and VARCHAR2 for names.
- EP_STUDENTS: Contains student profiles with a foreign key reference to their respective batches.
- EP_ATTENDANCE: Records daily presence/absence using constraints to ensure only 'Present' or 'Absent' values are stored.
- EP_ASSESSMENTS: Stores numerical scores (0-100) linked to student IDs to track academic achievement.




4. Data Population

Data was populated using optimized SQL Scripts to ensure consistency across the application. We inserted records for multiple students across different batches (Batch-A and Batch-B) to provide a sufficient dataset for meaningful analysis. This automated script approach ensures that the environment can be reset or migrated easily during the hackathon evaluation.

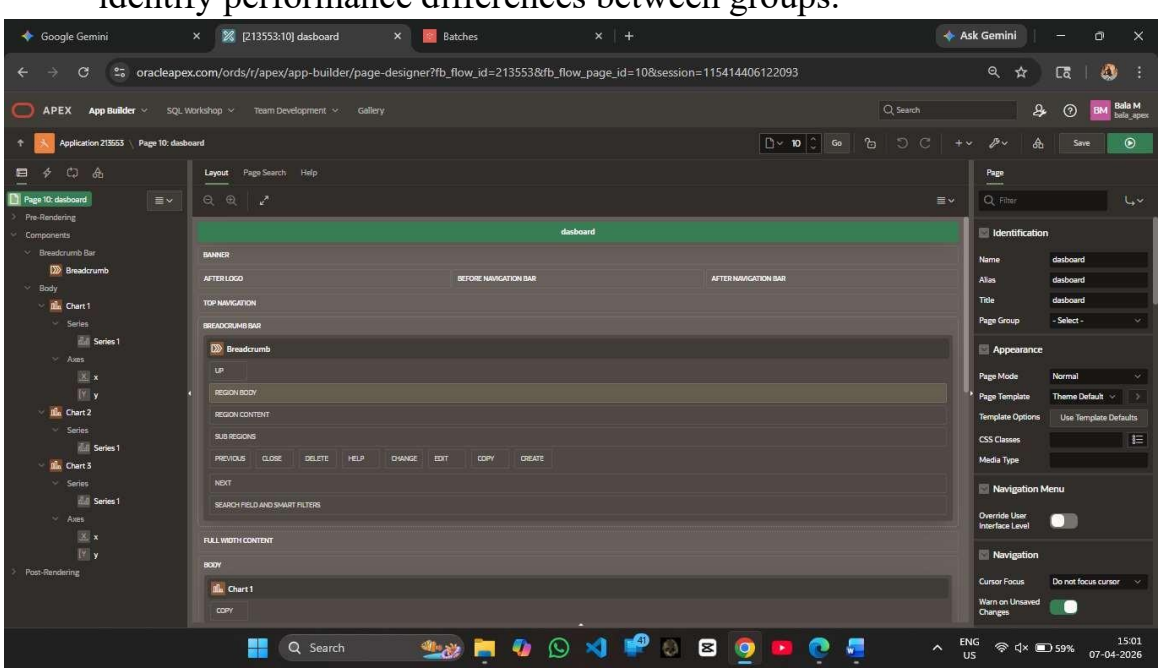



5. SQL Queries for Analysis

To drive the dashboard, specialized SQL queries and Database Views were created:

Performance Analysis: SELECT full_name, score FROM v_ep_performance to compare individual student results.

- Attendance Calculation: Aggregation queries like SELECT status, COUNT(*) FROM ep_attendance GROUP BY status to find the total headcount for presence and absence.
- Batch Comparison: Queries joining batch and assessment tables to identify performance differences between groups.



6. Reports Development

We developed Interactive Reports in Oracle APEX to allow users to manipulate data on the fly. These reports include advanced features such as:

- Column Sorting: To quickly find the highest and lowest scores.
- Filtering & Searching: To isolate specific student records or batches.
- Data Export: Capability to download report data for external use.

7. Data Visualization (Charts)

Visual storytelling is achieved through three distinct chart types:

- Bar Chart: Used for Score Comparison, showing how each student performs relative to their peers.
- Donut Chart: Provides a clear view of Attendance Distribution, highlighting the ratio of students present versus those absent.
- Line Chart: Tracks Performance Trends over time, helping educators see if class averages are improving or declining.

8. Dashboard Design

The dashboard is designed with a User-Centric Layout:

- Page Layout: Charts are placed at the top for immediate visual impact, followed by detailed reports.
- Navigation: A centralized Home Page with navigation cards ensures users can move logically between the Dashboard, Student Reports, and Batch Information.

9. Interactivity Features

The app features high levels of interactivity using Page Items and Dynamic Actions:

- Batch-wise Filters: A Select List allows users to filter the entire dashboard by a specific batch.
- Dynamic Refresh: When a filter is changed, the charts update instantly without a full page reload, providing a smooth user experience.

10. Best Practices Followed

The implementation adheres to industry standards:

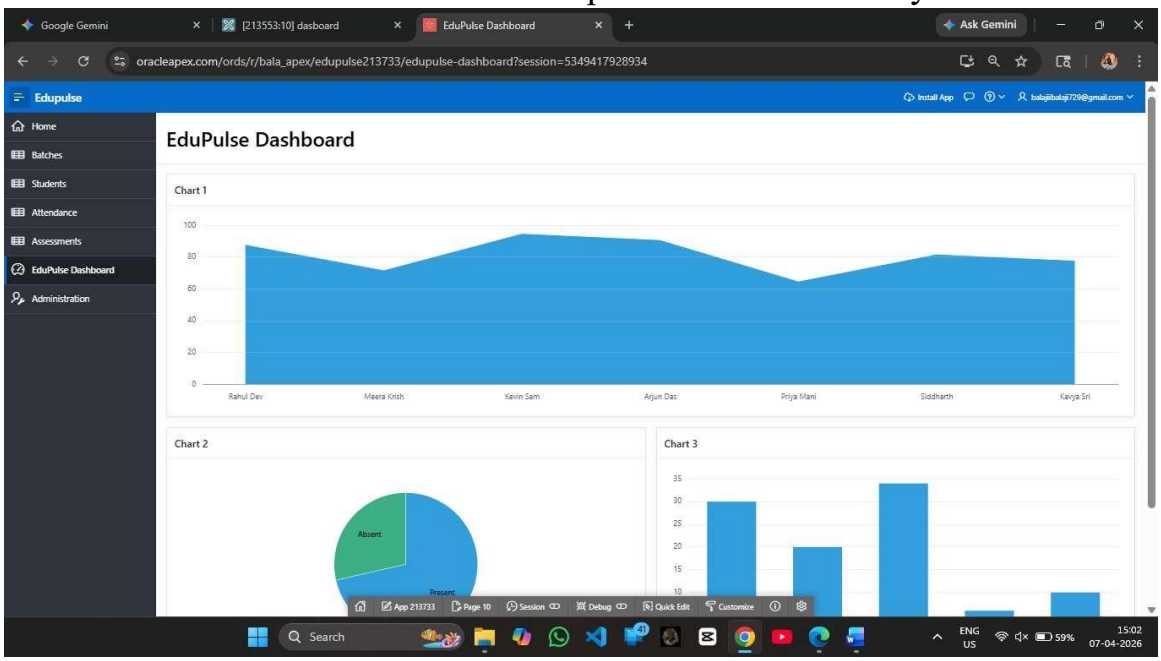
- Optimized SQL: Using Views to simplify complex joins for faster chart loading.
- Naming Conventions: Consistent use of prefixes (e.g., EP_) for all database objects.
- Responsive UI: Ensuring the dashboard is readable on various screen sizes.

11. Key Insights / Analysis

Through the EduPulse Dashboard, we identified:

- High-Performers: Students consistently scoring above 90% in assessments.
- At-Risk Students: Students with low attendance trends (e.g., below 75%) who may need academic intervention.
- Stream Performance: Insights showing which academic streams (e.g., Material Science) are excelling. 12. Expected Output / Deliverables The final submission includes:

1. A fully Working Oracle APEX Application export.
2. A comprehensive SQL Script for database tables and sample data.
3. Interactive Dashboards with multiple visualizations andynamicfilters.




13. Learning Outcome

This project provided hands-on experience in:

- Full-Stack Low-Code Development: Building apps rapidly with Oracle APEX.
- Advanced SQL: Writing complex queries for data analytics.
- Data Visualization: Mastering chart configurations and dashboard aesthetics.

14. Conclusion

Level-2 of the EduPulse Insights project successfully bridges the gap between raw data and meaningful analysis. By implementing a robust data model and interactive visualizations, we have provided a tool that empowers educational institutions to make data-driven decisions to improve student outcomes.

Github link: https://github.com/bala7447/edupulse-apex-app